THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · NCERT-TYPE UNIT TEST SET 1

Chapters 10–13: **Light: Mirrors and Lenses, Keeping Time with the Skies, How Nature Works in Harmony, Our Home: Earth, a Unique Life-sustaining Planet** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. According to the first law of reflection, the incident ray, the reflected ray and the normal at the point of incidence: 1

(a) lie in the same plane **(b)** are at right angles to each other
(c) are always equal in length **(d)** are parallel

2. The apparent daily motion of the Sun from east to west is caused by the: 1

(a) revolution of the Earth around the Sun **(b)** rotation of the Earth on its axis
(c) rotation of the Sun **(d)** motion of the Moon

3. An ecosystem consists of: 1

(a) living and non-living components interacting with each other
(b) only humans
(c) only soil and water
(d) only plants and animals

4. The Earth is the only known planet that supports life because it has: 1

(a) no atmosphere **(b)** water, air and a suitable temperature
(c) only a Moon **(d)** rings

5. The reflection from a rough surface in which parallel rays are reflected in different directions is: 1

(a) total reflection **(b)** regular reflection
(c) refraction **(d)** diffuse reflection

6. The Moon takes about ___ to go around the Earth once. 1

(a) 365 days **(b)** 24 hours **(c)** 27 days **(d)** 7 days

7. A series of organisms in which each one is eaten by the next is a: 1

(a) food web **(b)** food chain **(c)** habitat **(d)** ecosystem

8. The most abundant gas in the Earth's atmosphere is: 1

(a) argon **(b)** carbon dioxide **(c)** oxygen **(d)** nitrogen

9. The relation between the focal length (f) and the radius of curvature (R) is: 1

(a) R = 2f **(b)** f = 2R **(c)** R = f / 2 **(d)** R = f

10. The calendar based on the motion of the Earth around the Sun is the: 1

(a) solar calendar **(b)** lunisolar calendar
(c) weekly calendar **(d)** lunar calendar

11. The process of conversion of dead matter into simple substances by decomposers is: 1

(a) photosynthesis **(b)** respiration **(c)** pollination **(d)** decomposition

12. The Earth's water that is available as fresh water for use is: 1

(a) all the ice **(b)** a small part of the total water
(c) all the oceans **(d)** most of the water

13. According to the second law of reflection: 1

(a) angles are always 90 degrees
(b) angle of incidence is double the angle of reflection
(c) angle of incidence equals angle of reflection
(d) angle of reflection is zero

14. The Earth completes one rotation on its axis in about: 1

(a) 27 days **(b)** 12 hours **(c)** 24 hours **(d)** 365 days

15. Which of the following is an abiotic component? 1

(a) earthworm **(b)** fungi **(c)** grass **(d)** sunlight

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. State the laws of reflection. 2

17. Why does the Sun appear to rise in the east and set in the west? 2

18. What is an ecosystem? Give two examples. 2

19. List the three main layers of the Earth. 2

20. What is lateral inversion? Why is 'AMBULANCE' written inverted on the front of the vehicle? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Draw in words the ray diagram of reflection at a plane mirror and mark i and r. 3

22. Describe an activity to make a simple sundial and use it to measure time. 3

23. Draw in words a pond ecosystem and name its biotic and abiotic components. 3

24. Describe the layers of the Earth with one feature of each. 3

OR

Differentiate between a concave and a convex mirror in three points. 3

25. Explain how the seasons are caused. 3

OR

Explain nitrogen fixation and its importance. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. State and explain the laws of reflection with a labelled description. Verify the laws using an activity. 5

OR

Describe how the Earth's rotation and revolution lead to day and night, years and seasons. 5

27. Describe the components of an ecosystem with examples and show how they are interdependent. 5

OR

Describe the interior of the Earth with the layers, their states and composition, and how scientists know about the interior. 5

END OF QUESTION PAPER — Total: 50 Marks

MASTERMIND